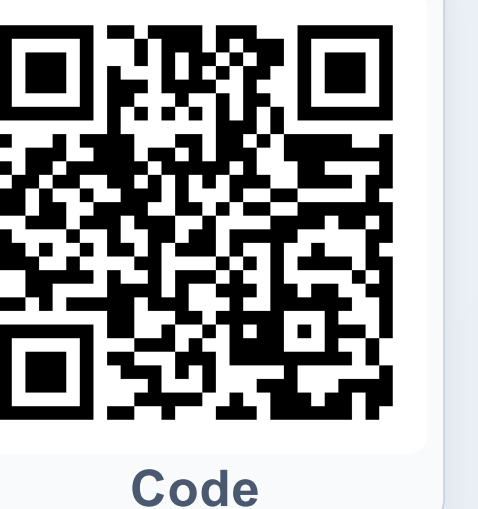
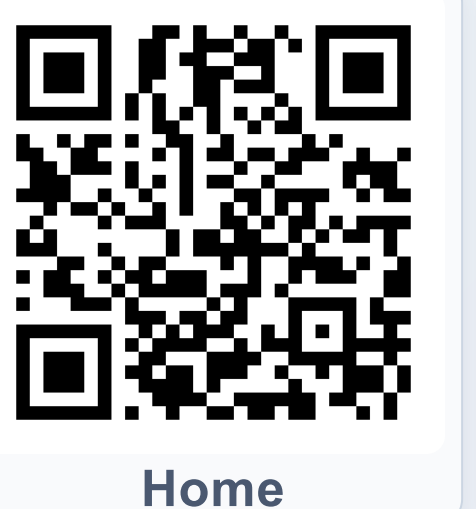




CMDs-AD: Cross-Modal Dual-Stream Decoupling for Few-Shot Anomaly Detection

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Problem

Few-shot multimodal anomaly detection must learn normality from only **1–4 paired RGB–3D normal samples**, too little support to cover ordinary changes in pose, illumination, texture, and geometry.

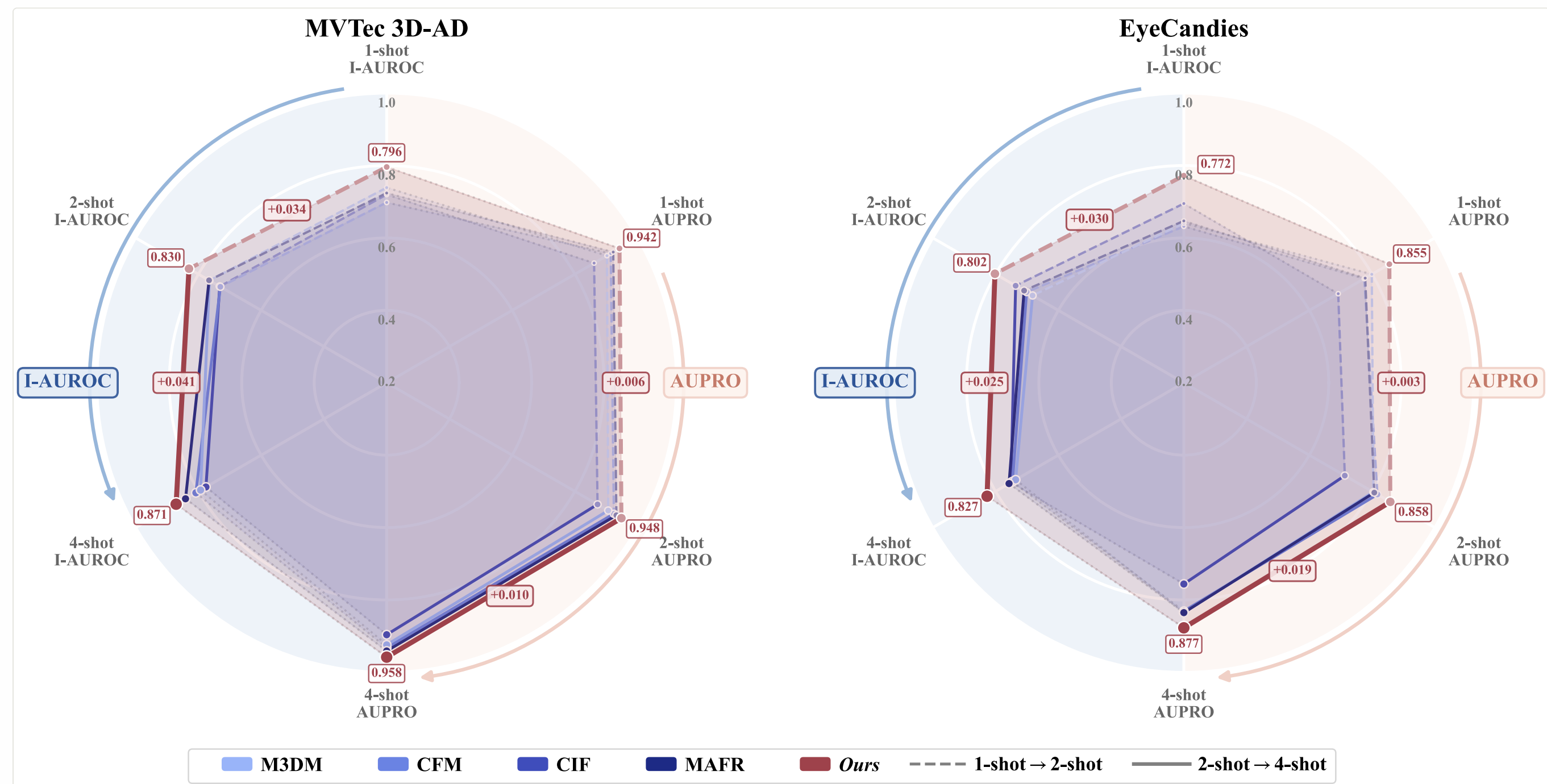
Uniform spatial processing mixes stable low-frequency structure with local high-frequency defects. Together with the RGB–geometry gap, this makes normal variation or sensor noise look anomalous and blurs the defect boundaries that inspection needs.

Motivation

Scarce normal coverage. One to four paired examples under-cover valid RGB–3D appearance and geometry.

Frequency entanglement. Tiny defects are high-frequency signals on stable low-frequency structure.

Unaligned modalities. RGB appearance and surface normals have different biases; naive fusion creates false positives.



CMDs-AD (red) scales from 1 to 4 normal samples on both benchmarks.

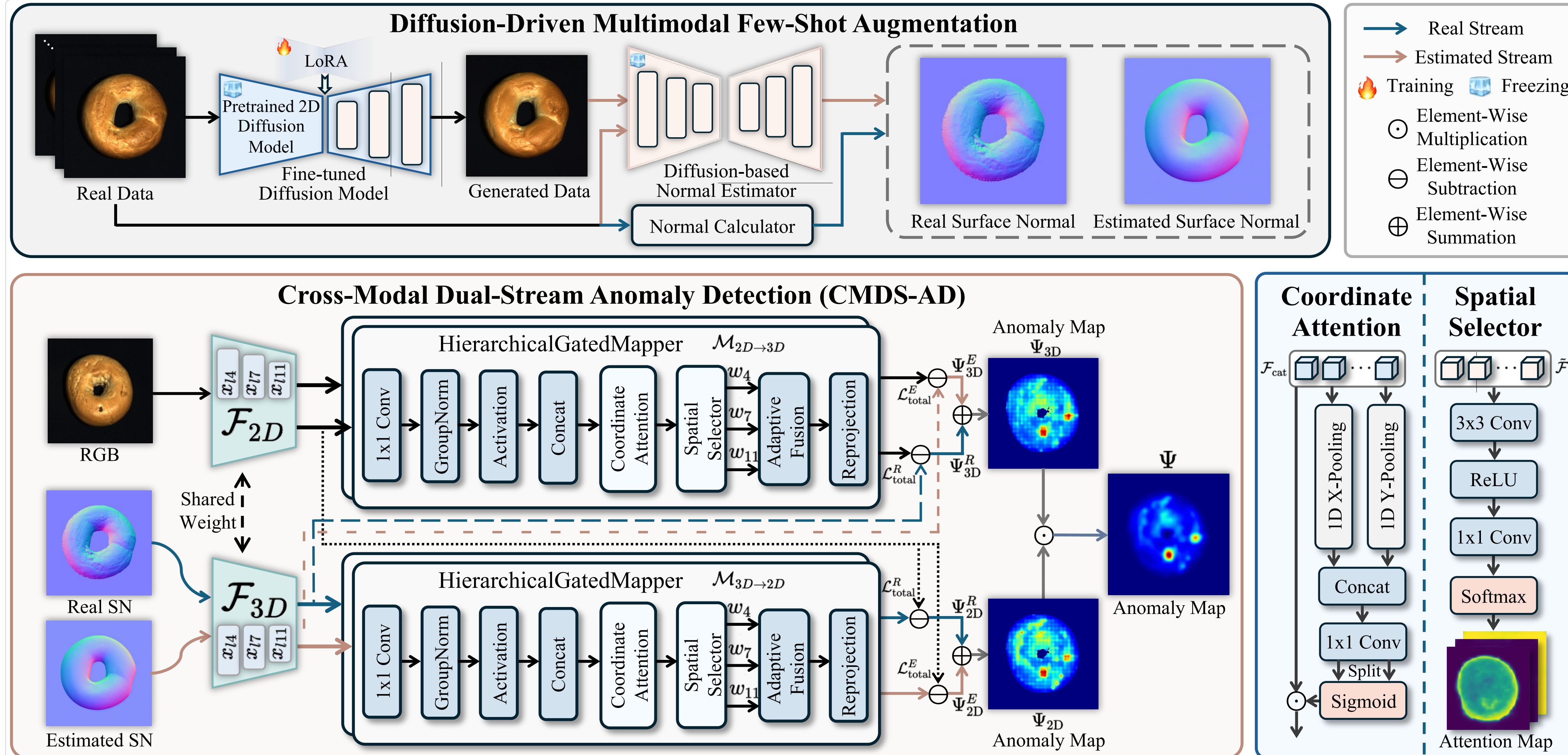
1 → 4 shot scaling	CMDs-AD
+7.5 MVTec 3D-AD · I–AUROC	+1.6 MVTec 3D-AD · AUPRO@30%
79.6 → 87.1 · +5.7 lead	94.2 → 95.8 · +2.0 lead
+5.5 EyeCandies · I–AUROC	+2.2 EyeCandies · AUPRO@30%
77.2 → 82.7 · +7.7 lead	85.5 → 87.7 · +5.6 lead

Readout. Across both benchmarks, paired RGB–3D performance improves as normal support grows from 1 to 4 shots.

Contributions

- Diffusion-driven dual stream.** LoRA adds paired RGB-normal views; estimated normals anchor structure while real normals preserve defect detail.
- Coordinate-aware alignment.** DINO layers 4/7/11 are selected and bidirectionally aligned with mask-aware cosine supervision.
- Strict anomaly scoring.** Real/estimated responses are multiplied across 2D and 3D to reject isolated noise.

Methodology



Paired diffusion augmentation creates real and estimated streams; bidirectional 2D↔3D mapping then fuses their dense anomaly evidence.

CMDs-AD: augment paired normality, decouple structure from detail, align RGB↔3D, then retain cross-modal agreement.

Dual-stream frequency role

$$\hat{N} = \Phi_{\text{diff}}(I) \approx N_{\text{low}}, \quad \mathcal{F}_{\text{est}} \approx \Phi(N_{\text{low}})$$
$$\mathcal{F}_{\text{real}} \approx \Phi(N_{\text{low}} + N_{\text{high}})$$

Separate structure from detail. Estimated normals anchor global shape; real normals retain high-frequency defects.

Coordinate-aware hierarchical alignment

$$\mathcal{L}_{\text{total}}^S = \alpha \mathcal{L}_{L4}^S + \beta \mathcal{L}_{L7}^S + \gamma \mathcal{L}_{L11}^S$$
$$S \in \{R, E\}$$

Map. Coordinate-aware projections preserve position before bidirectional 2D↔3D mapping; separate cosine losses supervise the real and estimated streams.

Balanced modality maps; strict fusion

$$\Psi_{2D} = \Psi_{2D}^R + \lambda_1 \Psi_{2D}^E, \quad \Psi_{3D} = \Psi_{3D}^R + \lambda_2 \Psi_{3D}^E$$
$$\Psi = \Psi_{2D} \odot \Psi_{3D}$$

Fuse only corroborated anomalies. Weighted streams form each map; the product keeps shared RGB–geometry evidence.

Experimental Details

Protocol. MVTec 3D-AD and EyeCandies; 1 / 2 / 4-shot paired RGB–3D normal samples.

Loss. Mask-aware averaging for the real stream; global averaging for the estimated stream.

Optimization. AdamW, 3,000 steps; $\alpha/\beta/\gamma = 1.2/1.0/0.8$; $\lambda_1 = \lambda_2 = 0.1$ for both streams.

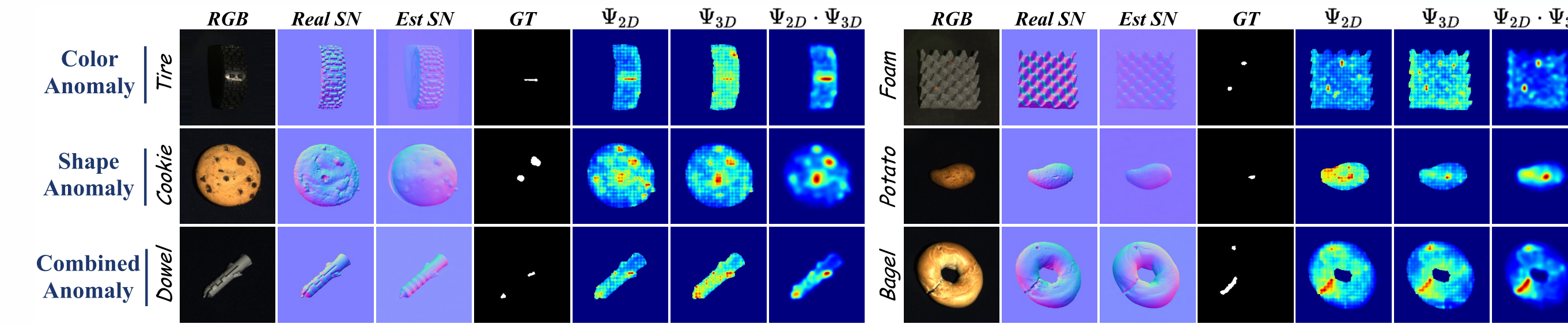
Mapping. Frozen DINO ViT-B/8 layers 4, 7, and 11 → 1×1 Conv–GN–GELU; coordinate + Softmax select 2D↔3D scales.

Quantitative Comparison

Five methods · 1 / 2 / 4-shot means

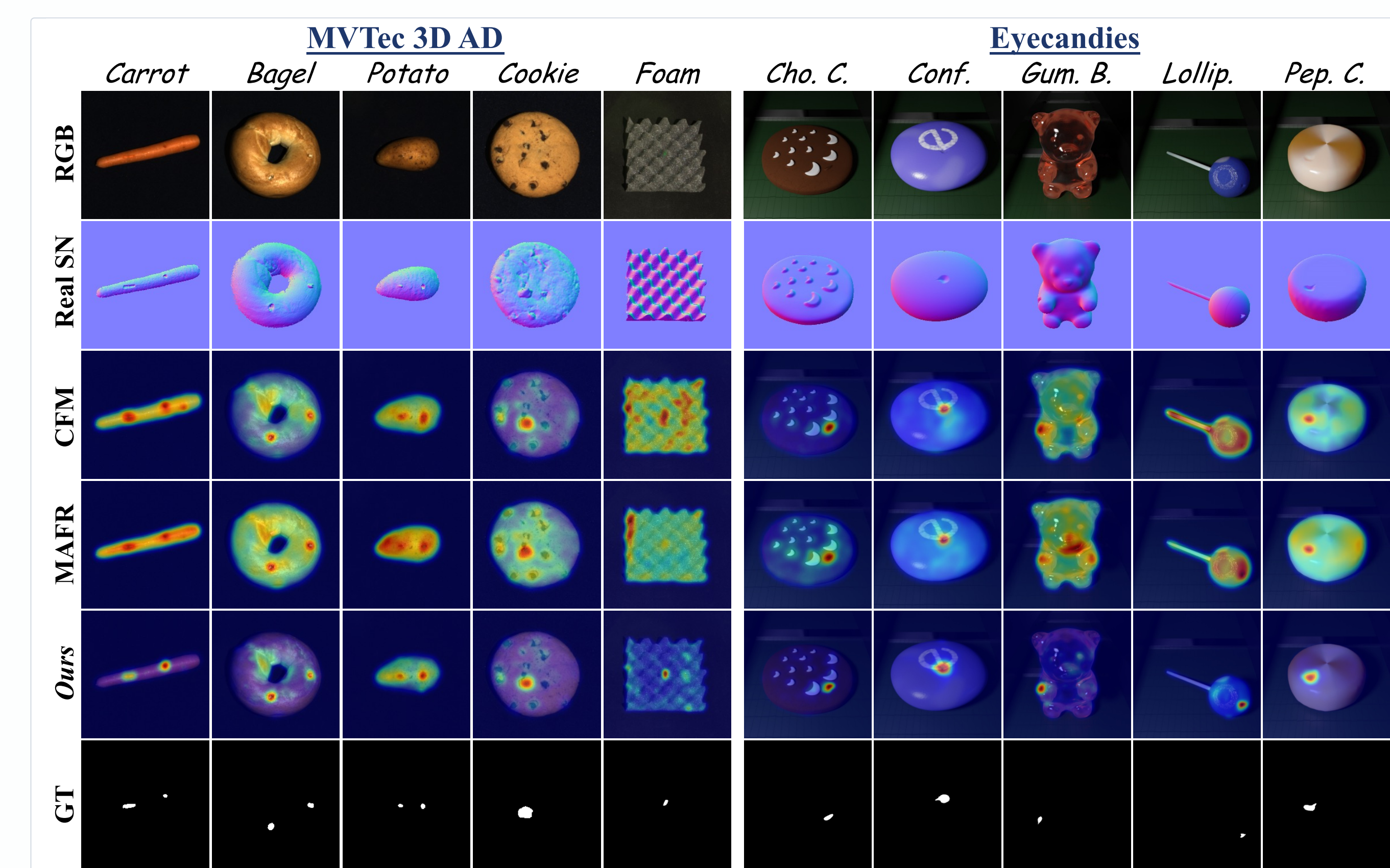
Method	MVTec 3D-AD 1 / 2 / 4 shot		EyeCandies 1 / 2 / 4 shot	
	I–AUROC	AUPRO@30%	I–AUROC	AUPRO@30%
CIF	72.0 / 73.2 / 77.6	86.1 / 87.2 / 89.6	69.5 / 73.6 / 75.1	69.2 / 71.3 / 75.6
M3DM	73.9 / 76.5 / 79.3	90.2 / 90.6 / 92.4	64.1 / 68.2 / 73.6	79.9 / 80.4 / 83.5
CFM	69.8 / 73.0 / 80.8	91.4 / 92.4 / 93.2	63.2 / 70.1 / 74.6	77.5 / 81.7 / 83.1
MAFR	72.4 / 76.6 / 84.1	92.2 / 93.2 / 94.1	64.7 / 70.9 / 75.7	77.8 / 80.7 / 83.5
CMDs-AD	79.6 / 83.0 / 87.1	94.2 / 94.8 / 95.8	77.2 / 80.2 / 82.7	85.5 / 85.8 / 87.7

2D / 3D / Fused Evidence



Color-only, shape-only, and combined defects are covered by complementary modality responses; the fused map suppresses unsupported background activation.

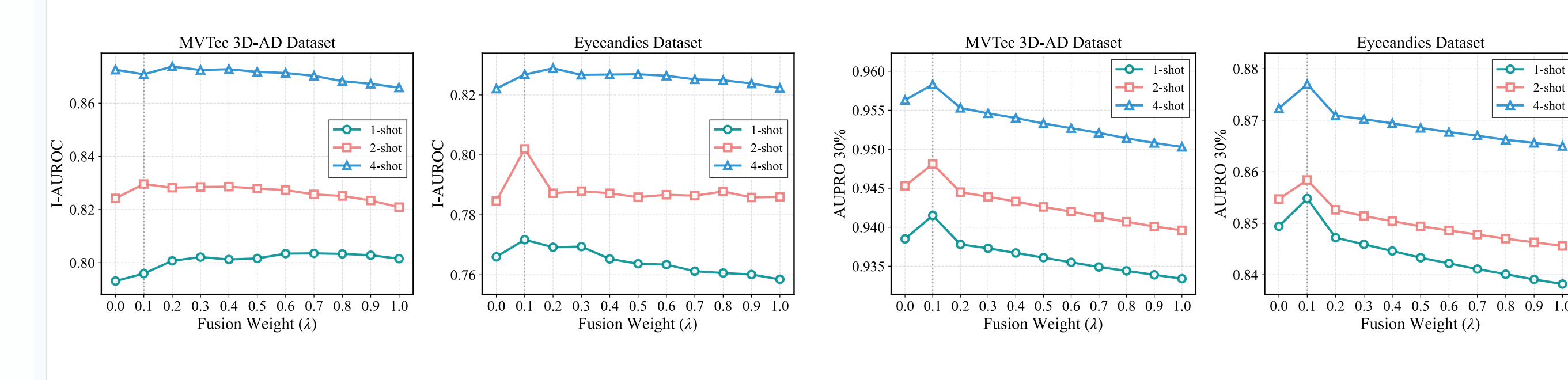
Cross-Method Qualitative Comparison



Full comparison: CMDs-AD localizes defects with fewer unsupported activations across both benchmarks.

Parameter Sensitivity

Stable around $\lambda_1 = \lambda_2 = 0.1$.



Performance remains stable around the selected balancing weights, with only small changes across the tested λ_1, λ_2 settings.

Ablation Study

4-shot MVTec 3D-AD · N = synthetic images per real image

VARIANT	I–AUROC	P–AUROC	AUPRO@30%	AUPRO@10%	AUPRO@5%	AUPRO@1%
Stability & synthesis						
5 seeds	86.8 _{±1.2}	98.9 _{±0.1}	95.9 _{±0.3}	88.8 _{±0.7}	80.9 _{±0.9}	41.3 _{±0.8}
N = 1	86.5	98.9	95.8	88.5	80.3	40.4
N = 3	86.2	98.9	95.9	88.7	80.5	40.6
Ours (N = 5)	87.1	98.9	95.8	88.6	80.6	41.0
Core components						
w/o feature mapper	87.2	98.8	95.5	87.7	79.2	39.9
w/o estimated stream	87.3	98.9	95.6	88.0	79.5	40.0
w/o real stream	81.3	97.9	93.2	83.1	73.2	35.0
w/ dual stream + mapper	87.1	98.9	95.8	88.6	80.6	41.0
Fusion strategies						
2D only · Ψ_{2D}	80.7	98.5	94.8	86.4	77.3	37.2
w/ 2D stream · Ψ_{3D}	84.8	98.4	94.5	85.5	76.6	38.1
w/ additive fusion · $\Psi_{2D} + \Psi_{3D}$	87.5	98.6	95.2	87.4	79.1	40.1
w/ max fusion · $\max(\Psi_{2D}, \Psi_{3D})$	86.1	98.5	95.0	86.9	78.5	39.4
w/ product fusion · $\Psi_{2D} \odot \Psi_{3D}$	87.1	98.9	95.8	88.6	80.6	41.0